SPINNING WHEELS & DRIVING MOTORS!

Control DC Motors with Arduino Uno & L298N Motor Driver

🧰 Materials Needed for this Lesson:

- 💻 1. Arduino Uno
- 🐌 2. L298N Module
- ⚙️ 3. DC Motor
- 🐢 4. 9V Battery
- 👃 5. Connecting Wires

1. Meet Your Robot's Core Components

To make a mobile robot or automated wheel spin, we use three fundamental hardware components working together:

- **The Brain (Arduino Uno):** Processes logic, executes programmed code, and sends low-power directional control signals.
- **The Translator (L298N Motor Driver Module):** Acts as a high-power bridge/switch between the Arduino signals and high-current motors.
- **The Muscle (DC Motor):** Converts electrical energy into mechanical rotational movement to spin wheels.

⚠️ CRITICAL SAFETY RULE: DIRECT CONNECTION RISK!
NEVER connect a DC motor directly to Arduino digital output pins. DC motors draw significantly more current than Arduino pins can safely provide (max 40mA), which will cause permanent hardware damage to your microcontroller board.

2. Hardware Wiring & Connections Guide

Follow this summary connection table to interface your external 9V battery, L298N driver module, Arduino Uno, and DC motor:

From Component	To Component	Purpose / Description
Arduino Digital Pin 8	**L298N IN1**	Direction Control Signal Line 1
Arduino Digital Pin 9	**L298N IN2**	Direction Control Signal Line 2
Arduino GND Pin	**L298N GND Terminal**	Common Ground Reference (Mandatory)
9V Battery Positive (+)	**L298N VCC (+12V/VCC)**	Main External Motor Power Source
9V Battery Negative (-)	**L298N GND Terminal**	Power Supply Ground Return
L298N OUT1 & OUT2	**DC Motor Terminals**	Power Output to drive motor windings

Think & Check — Why Common Ground (GND)? Connecting Arduino GND to L298N GND ensures both devices share a common reference voltage. Without a shared ground connection, control signals (HIGH/LOW) cannot be properly interpreted by the driver module!

3. Detailed Assembly & Wiring Steps

Follow these step-by-step instructions carefully to build your motor control circuit safely:

Step 1: Wire the DC Motor Terminals to L298N

Take the two leads from your DC Motor and insert them into the screw terminals marked **OUT1** and **OUT2** on the L298N module. Tighten the terminal screws using a small screwdriver so the wires are securely held.

Step 2: Connect the Control Signals (Arduino → L298N)

Using jumper wires, connect **Digital Pin 8** on the Arduino to **IN1** on the L298N. Next, connect **Digital Pin 9** on the Arduino to **IN2** on the L298N module. These two lines will pass forward/reverse commands to the motor driver.

Step 3: Establish a Common Ground Connection

Connect a jumper wire from any **GND pin** on the Arduino Uno directly into the **GND screw terminal** on the L298N module. This shared ground is essential for control signal stability.

Step 4: Connect the External Power Supply (9V Battery)

Ensure the battery snap connector is ready. Connect the **Positive (+ / Red) lead** of the 9V battery to the **+12V / VCC screw terminal** on the L298N. Connect the **Negative (- / Black) lead** of the 9V battery to the **GND terminal** on the L298N (sharing the same GND terminal with Arduino).

Step 5: Pre-Power Visual & Safety Inspection

Before connecting the battery or plugging in USB power, double-check that no loose wire strands are shorting adjacent terminals, that polarity is correct, and that ground wires are firmly attached.

4. Logic & Arduino Programming

The L298N driver uses logic combinations across control pins **IN1** and **IN2** to determine direction and braking:

IN1 (Pin 8)	IN2 (Pin 9)	Motor Action	Real-World Analogy
HIGH	**LOW**	Spins Forward (Clockwise)	Driving Forward
LOW	**HIGH**	Spins Backward (Counter-Clockwise)	Reversing / Backing Up
LOW	**LOW**	Stops Moving (Freewheel)	Neutral / Idle Coasting
HIGH	**HIGH**	Stops Moving (Brake)	Active Brakes Engaged

Sample Code: Wiper / Reversing Pattern Program

```
// Arduino Motor Control Code Example
void setup() {
  pinMode(8, OUTPUT); // L298N IN1 Direction Control
  pinMode(9, OUTPUT); // L298N IN2 Direction Control
}

void loop() {
  // 1. Spin Clockwise (Forward) for 2 Seconds
  digitalWrite(8, HIGH);
  digitalWrite(9, LOW);
  delay(2000);

  // 2. Stop for 1 Second
  digitalWrite(8, LOW);
  digitalWrite(9, LOW);
  delay(1000);

  // 3. Spin Counter-Clockwise (Backward) for 2 Seconds
  digitalWrite(8, LOW);
  digitalWrite(9, HIGH);
  delay(2000);

  // 4. Stop for 1 Second before repeating
  digitalWrite(8, LOW);
  digitalWrite(9, LOW);
  delay(1000);
}
```

5. Troubleshooting Checklist

Issue / Problem	Troubleshooting Steps & Solution
Motor Not Spinning At All	• Check if the external 9V battery is connected firmly to L298N VCC (+12V) and GND. • Verify Arduino GND is connected to L298N GND. • Ensure the power LED on the L298N board lights up. • Check battery voltage with a multimeter to ensure it isn't depleted.
Motor Spins in Wrong Direction	• Swap the **OUT1** and **OUT2** wire leads connected to the DC motor. • Alternatively, flip the `HIGH` and `LOW` signal assignments in your Arduino code.
Arduino Resets when Motor Starts	• Indicates motor is drawing current directly from Arduino power pin instead of external 9V battery. • Ensure 9V battery power is wired to L298N power terminal, not Arduino 5V.

Student Practice Sheet — Test Your Knowledge

Fill in your answers below or in your robotics lab notebook.

Name: ______________________________________ **Date:** ____________________

Q1. Primary Role

What is the primary role of the Arduino Uno in a robotic motor circuit?

Q2. Circuit Safety

Why is it unsafe to connect a DC motor directly to the digital output pins of an Arduino board?

Q3. Motor Driver Function

Explain the main function of the L298N Motor Driver module in your robot circuit.

Q4. Common Ground

What happens if you forget to connect the Arduino's GND pin to the L298N's GND terminal?

Q5. Motor Direction Logic State

Fill in the missing pin states for standard motor operation using Pin 8 (IN1) and Pin 9 (IN2):

```
• To rotate Forward:     Pin 8 = _______________ , Pin 9 = _______________
• To rotate Backward:    Pin 8 = _______________ , Pin 9 = _______________
• To Stop movement:      Pin 8 = _______________ , Pin 9 = _______________
```

Q6. Arduino C++ Syntax

What command in Arduino C++ code is used to create time delays between motor direction changes, and what unit of time does it measure?

Q7. Direction Troubleshooting

If your motor turns backwards when your code sends a Forward signal, what are two practical ways you can fix this problem?